

GAYATHRI TIRUMALA

Software Engineer

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PROFESSIONAL SUMMARY

Software Engineer with hands-on experience building Python applications, cloud-based data pipelines, and REST APIs. Experienced in Google Cloud Platform, SQL, and backend development with a strong foundation in scalable systems and data processing. Passionate about building reliable software, improving system efficiency, and solving real-world problems through clean, maintainable code

TECHNICAL SKILLS

- **Languages:** Python, Java, SQL, JavaScript, HTML, CSS
- **Backend & Frameworks:** Spring Boot, REST APIs, React (basic)
- **Cloud & Data:** Google Cloud Platform (BigQuery, Dataflow, Pub/Sub, Cloud Storage), AWS
- **Data & Libraries:** Pandas, NumPy, Matplotlib, Seaborn
- **Databases:** MySQL
- **Tools:** Git, Linux, Jira, Salesforce, Agile/Scrum
- **Concepts:** OOP, Data Structures, SDLC, ETL Pipelines, API Development, Unit Testing, Debugging

EDUCATION

Master of Science in Computer Science

University of Central Missouri | GPA: 3.7 | Jan 2024 – May 2025

Relevant Coursework: Advanced Algorithms, Advanced Operating Systems, Cloud Computing, Advanced Database Systems, Software Testing & Quality Assurance

Bachelor of Technology in Computer Science & Engineering

IARE Hyderabad | CGPA: 8.44 | Aug 2019 – Jun 2023

PROFESSIONAL EXPERIENCE

Graduate Software Engineer | University of Central Missouri | Kansas, USA Jan 2024 – May 2025

- Designed and implemented a secure cloud data-sharing system using Attribute-Based Encryption and Python to enforce fine-grained access control
- Implemented Python simulations to evaluate encryption performance and compare access control models
- Developed a full stack doctor appointment booking application with authentication, scheduling logic, and REST APIs
- Built backend services for an online course registration platform to manage users, schedules, and catalogs
- Implemented validation logic to prevent scheduling conflicts and improve system reliability
- Applied machine learning and data analysis techniques to study AI driven threat detection patterns
- Tested and validated applications handling thousands of simulated user records and datasets
- Collaborated in Agile team environments using Git, code reviews, and technical documentation

SOFTWARE ENGINEER | Client: Verizon Communications | Hyderabad, India Feb 2022 – Dec 2023

- Engineered Python ETL pipelines to process large customer datasets, reducing manual data preparation time by approximately 60%
- Built scalable data processing workflows using Google Cloud Dataflow and BigQuery
- Wrote complex SQL queries to analyze large datasets and support business reporting needs
- Automated data ingestion and archival using Google Cloud Storage and Pub/Sub
- Created data visualizations using Matplotlib and Seaborn to communicate insights to stakeholders
- Improved pipeline performance and reliability through testing, debugging, and code reviews
- Optimized data pipeline scalability to handle millions of records using GCP Dataflow
- Collaborated in Agile teams, participating in sprint planning, daily standups, and peer code reviews

SOFTWARE ENGINEER INTERN | INCEDO INC | Hyderabad, India Aug 2021– Jan 2022

- Built Python automation scripts to improve data validation and reduce manual effort
- Performed exploratory data analysis using Pandas and SQL to identify trends and anomalies
- Developed cloud-based workflows using Google Cloud Storage and BigQuery
- Wrote technical documentation for data pipelines and validation processes

CERTIFICATIONS

- Google Cloud Associate Cloud Engineer (ACE)